Two Dimensional, Transient Heat Transfer
Analytical Solution

1. Introduction

In most realistic cases involving heat transfer, analytical solutions to the temperature distribution within the domain of interest are impossible to derive without making simplifying assumptions or otherwise condensing the problem in some manner. The labs we have considered in MEEG346 have allowed for analytical solutions by either neglecting heat transfer in various dimensions or by ignoring some of the physical phenomena associated with the heat transfer process. Here we will examine a more complex analytical solution for 2D heat transfer in order to illustrate the complexity of some of these mathematical solutions.

1.1 Lab Objectives

In this lab you will examine analytical solutions to heat transfer problems and compare the results with measured and simulated data. During the course of this lab you will:

- Observe how analytical solutions to heat transfer problems can be derived.
- Plot solutions to these analytical problems.
- Compare and contrast the results of your analytical solution with measured and simulated datasets.

The background theory for this lab will be developed in your heat transfer class. Additional information will be provided in your assigned lab session and on Canvas.

2. Solution of the 2D Temperature Distribution in the Thermochromic Plate

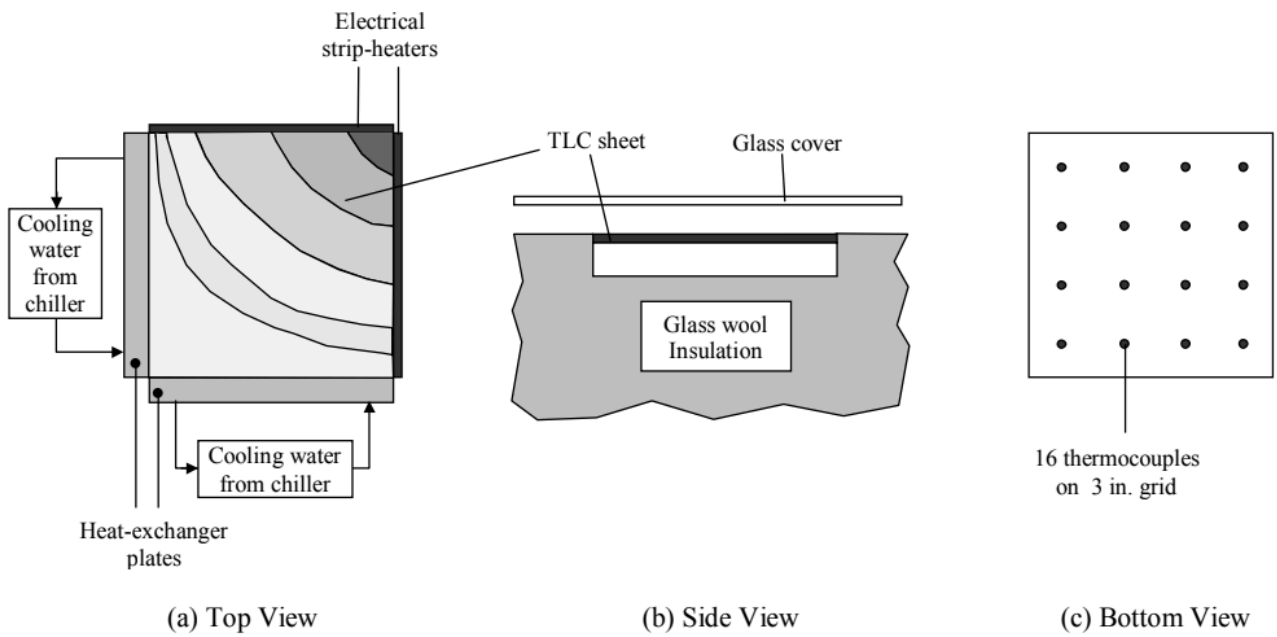
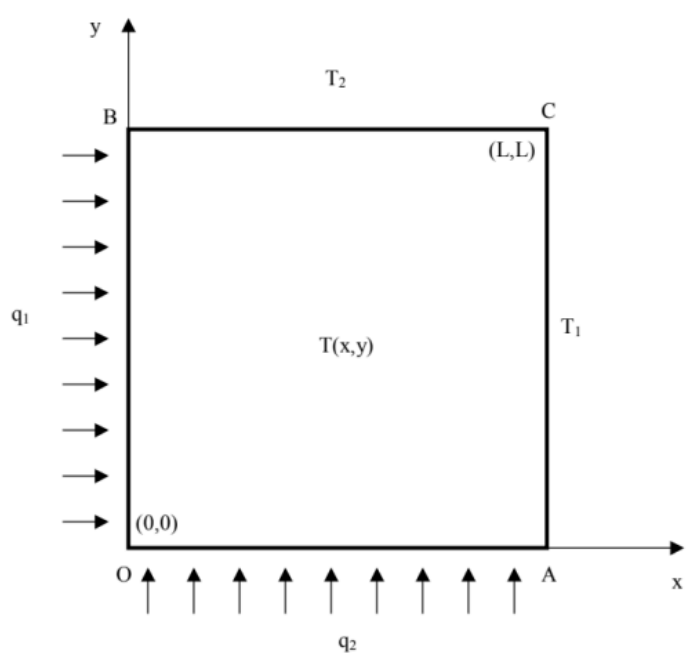


Figure 1: Schematic of heated plate



$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = \frac{1}{\alpha}\frac{\partial T}{\partial t}$$

$$L_x = L_y = L$$

Initial Condition

t=0 $T(x,y,0) = T_i$

Boundary Condition along the four sides

x=0 $q_1 = -k\dfrac{\partial T(0,y,t)}{\partial x}$

y=0 $q_2 = -k\dfrac{\partial T(x,0,t)}{\partial y}$

y=L $T(x,L,t) = T_2$

x=L $T(L,y,t) = T_1$

Figure 2: Mathematical model of experiment

Figure 1 details an overview of the 2D transient system that was analyzed experimentally and numerically as part of lab 3. Figure 2 breaks this experiment down into a mathematical model of the domain and boundary conditions and details the governing equations for this problem in terms of the temperature profile as a function of space and time.

The analytical solution of the steady state temperature profile within the system shown in Fig.2 is given by the equation 1, derived using the method of separation of variables (see Appendix A):

$$T(x,y) = T_1 + \frac{q_1''}{k}(L-x) + \sum_{n=0}^{\infty} \frac{2\cos(\lambda_n x)}{kL\lambda_n^2} \times \frac{\left[(T_2 - T_1)(-1)^n \lambda_n k - q_1''\right]\cosh(\lambda_n y) + q_2''(-1)^n \sinh\left[\lambda_n (L-y)\right]}{\cosh(\lambda_n L)}$$

$$\lambda_n \equiv \frac{(n+0.5)\pi}{L}$$

Equation 1: Steady state solution of 2D temperature profile

3. Procedure

Use the information provided in equation 1, as well as your knowledge of this experiment and measured data, to plot the steady state temperature profile at the conditions you measured in the lab. Develop a computer code to do this in either MATLAB or Python.

4. Data Analysis & Discussion

Provide an annotated copy of the derivation provided in Appendix A that details your understanding of how this solution was obtained.

Analyze your data and answer the following questions:

1. Plot your temperature contours obtained from the analytical solution.
2. Compare this profile with your measured and numerical data at steady state.

4.1 Comparison of Results

1. How well do the three sets of steady state data compare (thermocouples, analytical and numerical solutions)?
    a. Develop a series of plots to compare your results
    b. Consider overall comparisons and comparisons at a single point(s) on the plate using some form of image correlation.
2. How well does your data compare?

3. What are possible sources of error in the analytical model that might lead to a lack of correlation between the data sets?

5. Lab Deliverables

For this lab you should submit a technical document detailing your data, analysis, and discussion to Canvas for grading. This document should not be a full lab report, but should be professionally formatted using headers, figure captions and similar. Take care that your document addresses all the questions and items listed in Section 4. Data Analysis and Discussion. Your document should be uploaded as a PDF.

For this assignment you are also required to upload a functioning copy of your code. We should be able to download and run this code without needing to change any variables or similar. Upload this as a separate file to your PDF, do not zip your files.

Appendix A: Derivation of Steady State Temperature Distribution

Derivation courtesy of Dr. Lian-Ping Wang.

$$\partial_{xx}T + \partial_{yy}T = 0$$
$$BCs:\ -k\partial_x T(0,y) = q_1, \qquad T(L,y) = T_1,$$

$$BCs:\ -k\partial_y T(x,0) = q_2, \qquad T(x,L) = T_2,$$

The method is the separation of variables. We perform a transformation in the x direction to make two BCs homogeneous:

$$T(x,y) = \theta(x,y) + T_1 + \frac{q_1}{k}(L-x)$$

Then, in terms of $\theta(x,y)$, we have

$$\partial_{xx}\theta + \partial_{yy}\theta = 0$$

$$BCs:\ -k\partial_x\theta(0,y) = 0, \qquad \theta(L,y) = 0,$$

$$BCs:\ -k\partial_y\theta(x,0) = q_2, \qquad \theta(x,L) = T_2 - T_1 - \frac{q_1}{k}(L-x)$$

Let

$$\theta = X(x)Y(y)$$
$$\frac{X''}{X} = -\frac{Y''}{Y} = constant = -\lambda_n^2$$

In view of the first set of BCs. Then

$$X = \cos(\lambda_n x), \quad with\ \lambda_n = \frac{(n+0.5)\pi}{L}, n = 0,1,2,3, \ldots.$$

$$\frac{Y''}{Y} = \lambda_n^2, \rightarrow \quad Y = C_n\cosh(\lambda_n y) + D_n\sinh(\lambda_n y)$$

Then

$$\theta(x,y) = \sum_{n=0,1,2,\ldots}[C_n\cosh(\lambda_n y) + D_n\sinh(\lambda_n y)]\cos(\lambda_n x)$$

BCs:

$$k\sum_{n=0,1,2,\ldots}\lambda_n[D_n]\cos(\lambda_n x) = q_2$$

$$\sum_{n=0,1,2,\ldots}[C_n\cosh(\lambda_n L) + D_n\sinh(\lambda_n L)]\cos(\lambda_n x) = T_2 - T_1 - \frac{q_1}{k}(L-x)$$

Now use the orthogonal properties of the cosine function to determine C_n and D_n.